

Al-Azhar University
Faculty of Eng.,
Systems and Computer Eng., Dept.,

SYS 401: Embedded Systems

Grade 4, 2nd Semester

By: Ass., Prof Khaled Elshafey

Course Information

- Grading: Projects, Assignments, Exams
 - Assignments 5
 - Mid Term 15
 - Projects 10
 - Finals 70
- Labs
 - Microcontrollers, Sensors
- Projects
 - Project 1: Data Logger
 - Projects 2: Smart Buildings (Power saving)

Text Book

- - Introduction to Embedded Systems- A Cyber – Physical Systems Approach, by: Lee & Seshia, 2nd Edition, 2015.

Course Layout

- Introduction
- Modeling
- Sensors and Actuators
- Input and Output
- Multitasking and Scheduling
- Security and Privacy

Introduction

- The most visible use of computers & S.W is processing information for human consumption.
- The vast majority of computers in use, however, are much less visible.
- They run the Engine, Brake, Seatbelts, Air bags, Encode your voice and construct a radio signal to send it from your cell phone, ...etc.
- All hidden computers are called “ Embedded Systems”.

Contd.,

- Embedded Systems (E.S) can be defined as:
“A computer system with a dedicated function within a large mechanical or electrical Systems”.
- Another definition:
“Information processing systems with a dedicated function embedded into enclosing products, such as Cars, Telecommunication equipment,..”.

Types of ES's

- Embedded Hardware.

Is a combination of microchip and several electronic components which can able to perform some tasks.

- Embedded Software.

If you are using a controller, you have to code your needs (Firmware) to be able to perform the system operations.

Firmware

- The program instructions written for embedded systems are referred to as “Firmware”.
- Is a type of S.W that provides control, monitoring, and data manipulation of electronic products.

Common Characteristics of ES's

- ES's are connected to the physical environment through **Sensors & Actuators**.
- Have to be **Dependable**.
- Have to be **Efficient** (Energy, code-size, Run-time, Weight, and Cost).

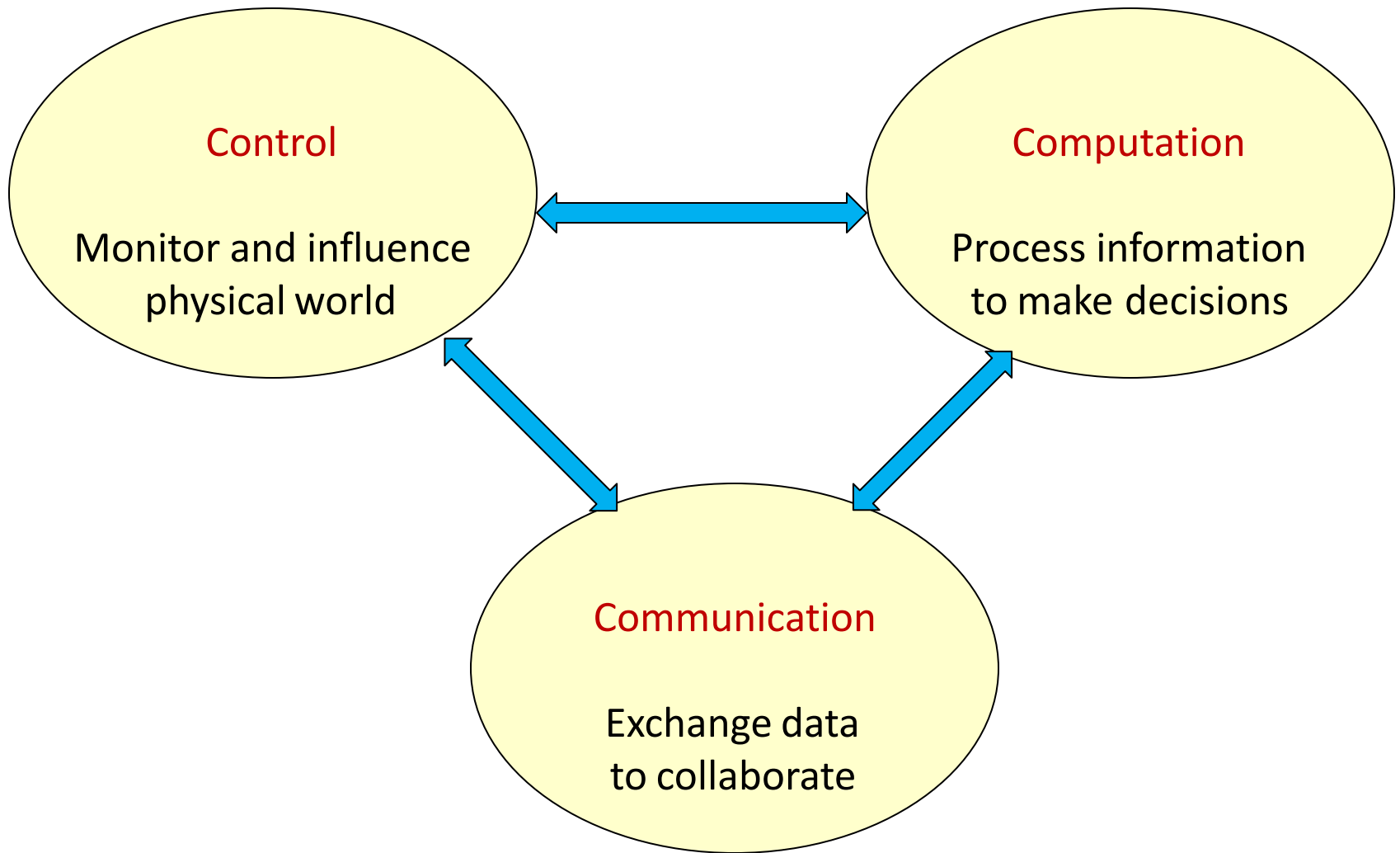
Contd.,

- **Dedicated** toward a certain application.
- Must meet “**Real-time constraints**”.
- Most ES's are **Hybrid** systems (Digital& Analogue parts).
- **Reactive** Systems (continual interaction with its environment).
- **Multitasking**. Many things happen at once.

Cyber-Physical Systems (CPS)

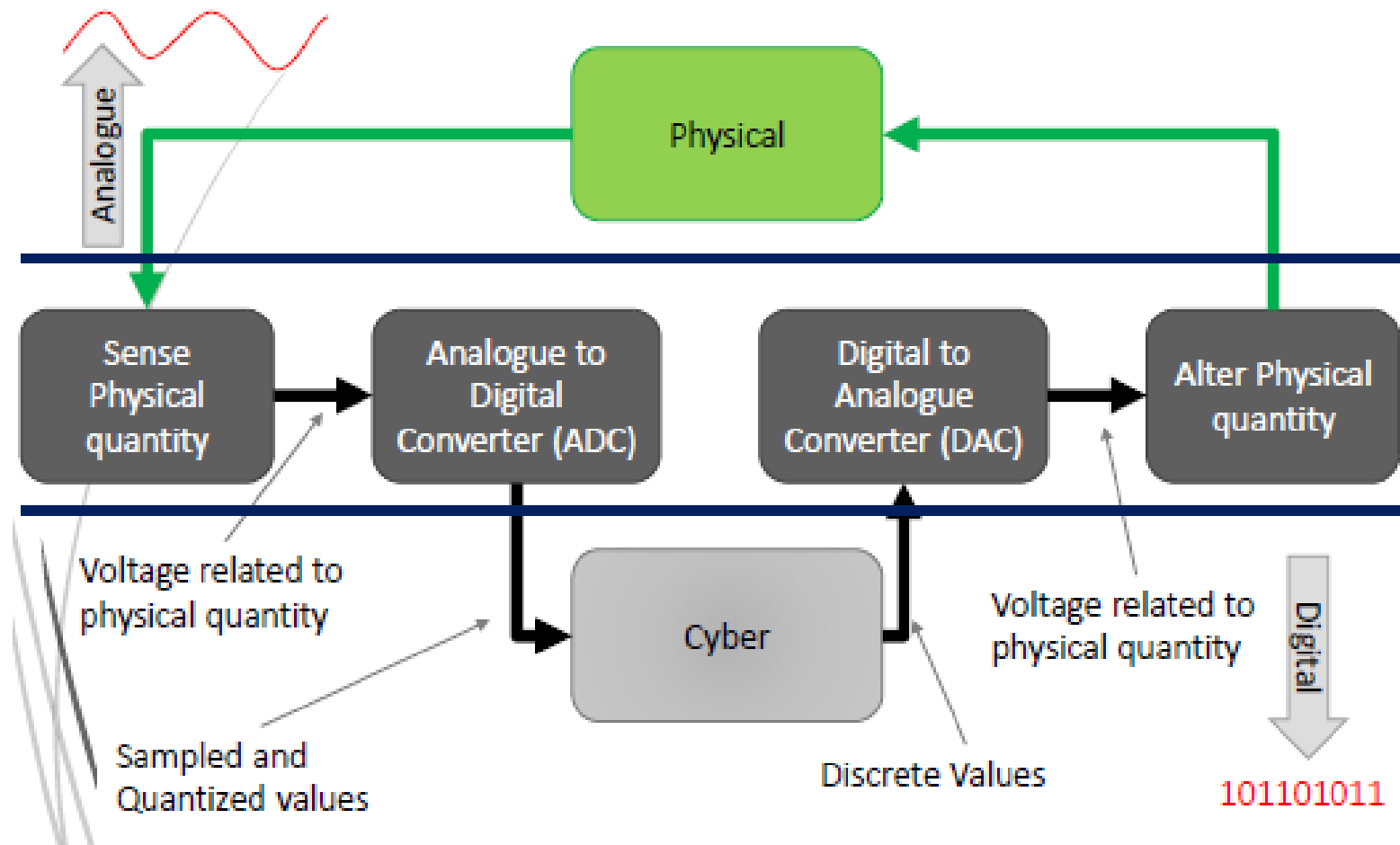
- CPS is an integration of computation with physical object processes.
- Embedded computers & Networks $\Rightarrow$ monitor and control the physical object processes usually with feedback loops.
- In CPS, many things happen at once.
- It uses smart sensors & Actuators

Cyber-Physical Systems

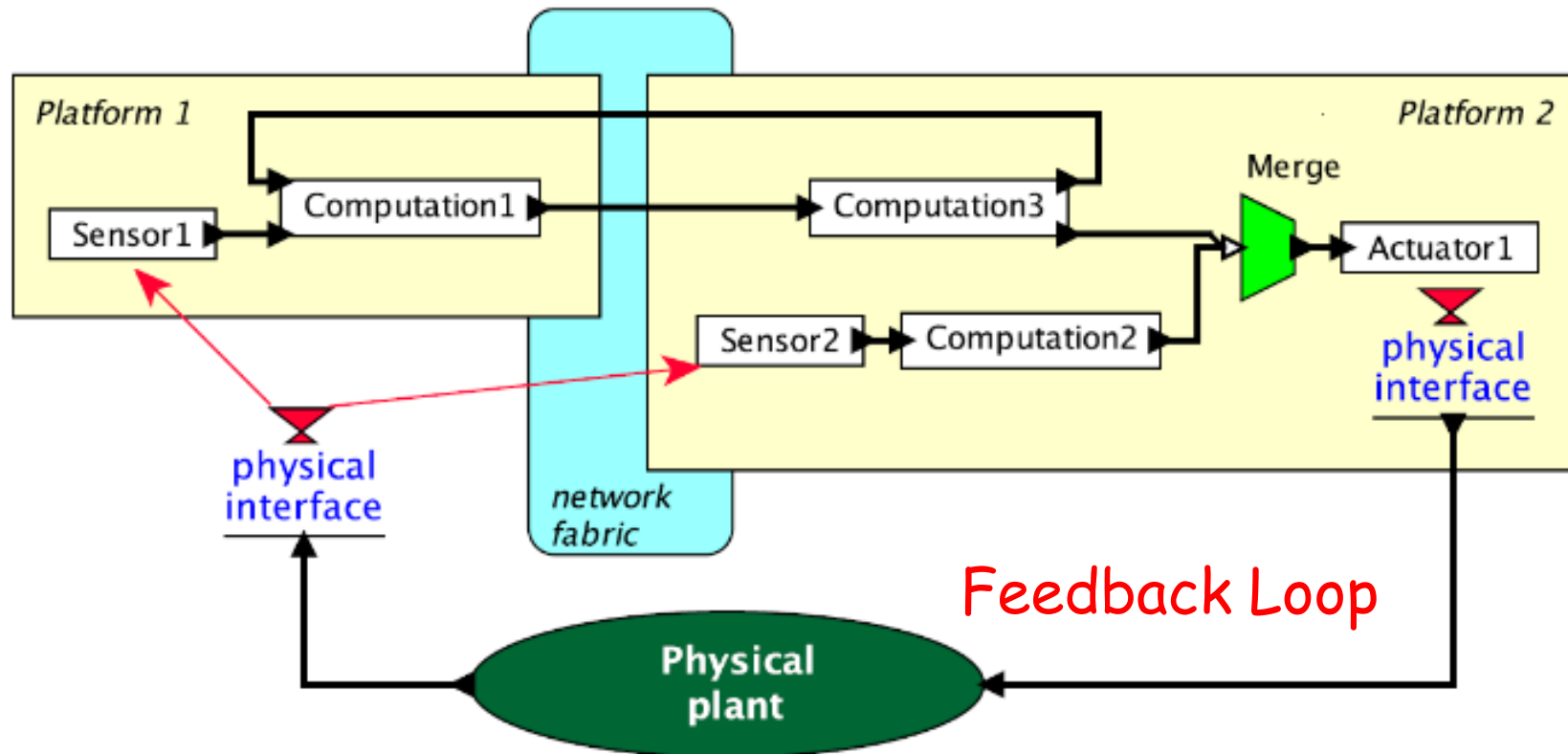


Cyber-Physical Systems

Bridges between physical and cyber worlds



CPS Structure Example



Internet of Things (IoT)

- Is the network of physical objects (provided with unique id) embedded with electronics, S.W, Sensors, and network connectivity, which enables these objects to collect and exchange data.
- The IoT allows objects to be sensed and controlled remotely across existing network infrastructure.
- It means : “ Network with smart devices”

IoT has Evolved from:

- Wireless Technologies
- Micro-electromechanical systems (MEMS)
- Internet

Motivating Example

- **STARMAC**

- Is a small quadrotor aircraft to serve as a testbed for experimenting with vehicle autonomous control techniques.
- The objective is to be able to have multiple vehicles cooperate on a common task.

Contd.,

- **The challenges are:**

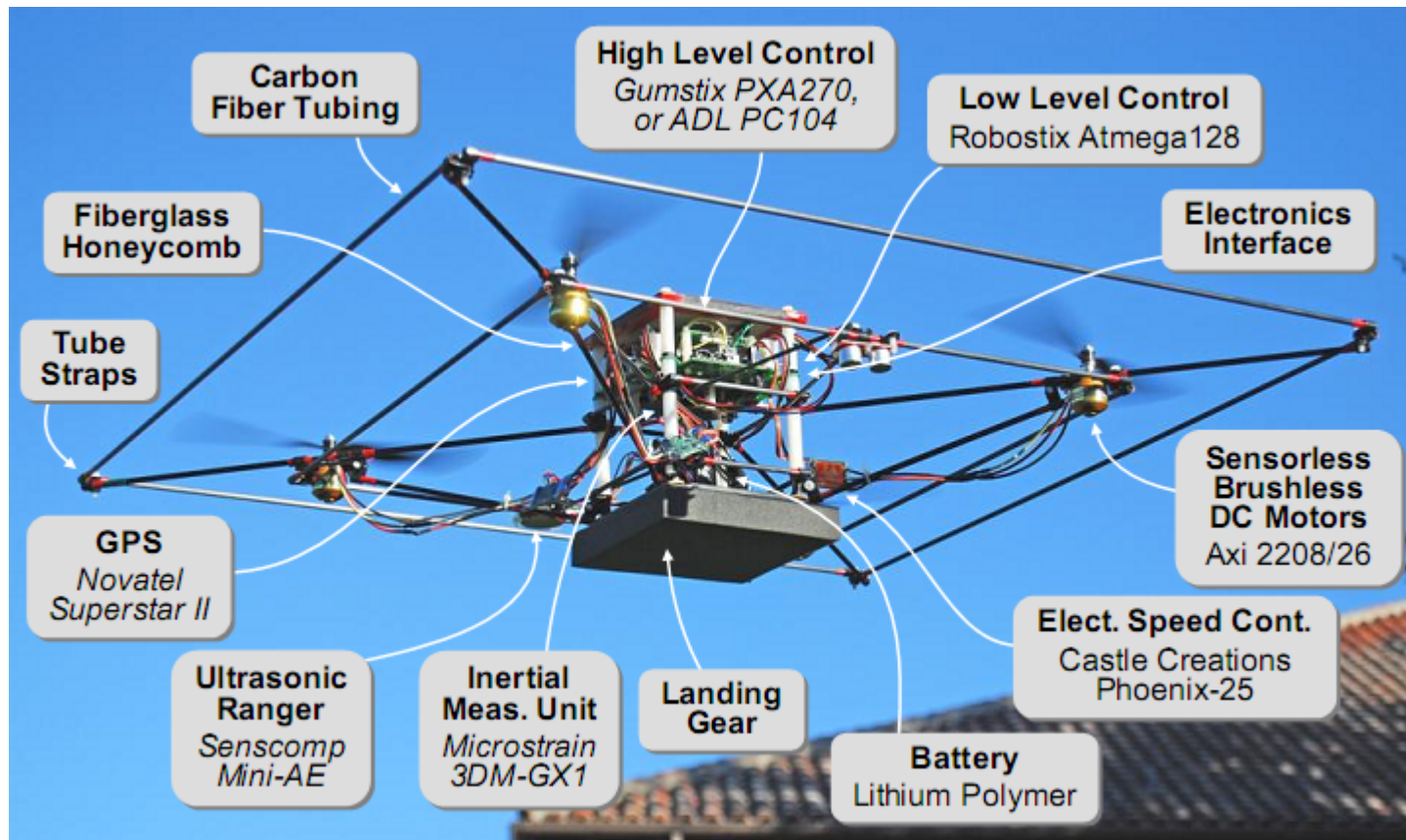
- 1- Controlling the vehicles

- 2-The main actuators are the four rotors which produces a variable amount of thrust.

- 3- The weight of the vehicle.

- 4- Interaction with its environment.

- 5- Self-Test



Design & Implementation Of CPS

- **Modeling** ? Is the process of gaining a deeper understanding of a system through imitation.
- **Design** ? Specifies how a system does what it does, by selecting components and putting them together.
- **Analysis** ? Every system must be designed to meet certain requirements.

Embedded Operating System

- An embedded operating system is a specialized OS for use in the computers built into larger systems.
- In contrast to an operating system for a [general-purpose computer](#), an embedded operating system is typically quite limited in terms of function – depending on the device in question, the system may only run a single application.
- However, that single application is crucial to the device's operation, so an embedded OS must be reliable and able to run with constraints on memory, size and processing power.

Contd.,

- Depending on the method used for multitasking, this type of OS is frequently considered to be a [real-time operating system](#), or *RTOS*.
- **RTOS** is an [operating system](#) (OS) intended to serve [real-time](#) applications that process data as it comes in, typically without buffer delays.
- Linux on ES, and Windows Embedded Automotive.

What Is a Real-Time System?

- A computer program/algorithm has inputs and outputs.
- The response/output to a given input defines the program behavior.
- In a real-time system, the behavior must be both logically correct and produced at the correct instant of time.
 - If the deadline is missed, the behavior is considered incorrect.

Hard vs. Soft Real-Time: Deadline

- A hard real-time system must produce the result at the correct physical time. Failure to do so can have catastrophic consequence and is considered unacceptable.
 - Nuclear power plant
- A soft real-time system can miss a deadline.
 - The result may still be useful
 - If the deadline is missed, there will not be a catastrophe. (e.g., multimedia)

Hard vs. Soft Real-Time: Response-Time

- A hard real-time system must respond to events on the order of milliseconds (e.g., a braking system).
- A soft real-time system's response can be on the order of seconds (an ATM machine).

Hard vs. Soft Real-Time: Peak-Time Performance

- A hard real-time system must meet the specified deadline under peak/worst conditions. It must be able to meet deadlines in rare-event scenarios.
- Soft real-time systems are defined in terms of average behavior. Performance degradation in worst case scenarios is accepted for economic reasons

Fail-Safe Vs. Fail-Operational

- If a system can enter a “safe” state when failure occurs, the system is fail-safe
 - Railway signaling system failure: a safe state is to turn all lights red.
 - Example: a watchdog timer or computer operating properly, must be stroked at regular intervals by software. If there is a software failure, the watchdog fires, forces the system into a safe state & resets the CPU.
- In a fail-operational system, a safe state after failure is not acceptable. The system must remain operational and provide a minimal amount of guaranteed service.
 - An airplane with a single failed engine

Application Areas

- Hybrid embedded systems
 - Aerospace, automobiles, robotics, process control, and sensor nets
- Multimedia
- Consumer electronics
 - Appliances, office electronics, and home/office automation
- Network components
 - Bridges, routers, switches, and hubs
- Medical instruments
 - Patient monitoring, MRI, and artificial hearts
- E-Business
 - ATM, vending machines
- Distributed & grid computing

What is a Microcontroller?

- A **microcontroller** is a single chip, self-contained computer which incorporates all the basic components of a personal computer on a much smaller scale.
- Microcontrollers are often referred to as single chip devices or single chip computers.
- The main consequence of the microcontroller's small size is that its resources are far more limited than those of a desktop personal computer.

microcontroller vs microprocessor

- A microprocessor is a single chip CPU used within other computer systems.
- A microcontroller is itself a single chip computer system.

A microcontroller main components:

A microcontroller has seven main components:

- Central processing unit (CPU)
- ROM
- RAM
- Input and Output
- Timer
- Interrupt circuitry
- Buses

The Microcontroller in a System

- Microcontrollers do not function in isolation.
- As their name suggests they are designed to control other devices.
- The microcontroller can accept inputs from some devices and provide outputs to other devices within any given system.
- For example, a microcontroller may accept input from a switch and may send output to an LED.

Microcontroller Architecture

- There are two basic types of architecture: Harvard and Von Neumann.
- Microcontrollers most often use a Harvard or a modified Harvard-based architecture.

Von Neumann

- Von Neumann architecture has a single, common memory space where both program instructions and data are stored.
- There is a single data bus which fetches both instructions and data.
- Each time the CPU fetches a program instruction it may have to perform one or more read/write operations to data memory space.
- It must wait until these subsequent operations are complete before it can fetch and decode the next program instruction.
- The advantage to this architecture lies in its simplicity and economy.

Harvard

- Harvard architecture computers have separate memory areas for program instructions and data.
- There are two or more internal data buses which allow simultaneous access to both instructions and data.
- The CPU fetches instructions on the program memory bus.
- If the fetched instruction requires an operation on data memory, the CPU can fetch the *next* program instruction while it uses the data bus for its data operation.
- This speeds up execution time at the cost of more hardware complexity.

I/O Ports

- There are two main port types, parallel and serial, and two port modes, synchronous and asynchronous.
- Parallel I/O requires a data line for each bit, while serial I/O uses a single line and transfers bits in sequence.
- Synchronous I/O is synchronized to a clock while asynchronous I/O is not.
- Microcontrollers most often have parallel I/O capability built in and serial I/O as a peripheral feature.

Assignment

- What are the differences between Embedded SW developer and SW developer.

Sheet

- Text Book Ch.,1